

1. Spatial KDA memory: architecture and causal tests

Technical research note · 23 September 2026

Scope: the current fresh-weight, 13-task joint learner, continued from step 715. Architecture is implemented; behavioral metrics are measured; all microstimulation protocols below are proposals, not completed interventions.

Executive assessment

The model combines a convolutional visual hierarchy with three spatial associative memories and a final global gated recurrent unit (GRU). Memory therefore affects visual processing at multiple resolutions, rather than existing only behind the encoder. Each of thirteen tasks has a separate linear classifier, but all visual processing, spatial memories and the global recurrent summary are shared. There is no explicit object-slot memory, no top-down GRU-to-encoder feedback, and no spatial softmax that competitively divides a fixed attention budget among scene locations.

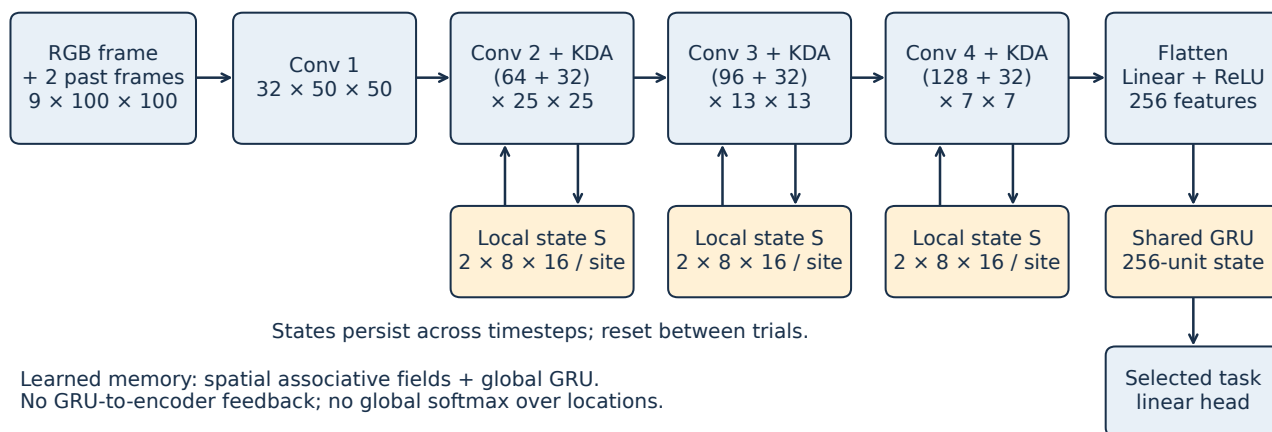


Figure 1: Implemented computation. Gold boxes are recurrent state, not separately trained modules. Every learned component remains trainable.

The completed short run acquired contrast, chromatic increments and natural-image spectral-detail discrimination, but not robust spatial selection, motion accumulation, binding or scene recognition. The ongoing continuation has substantially improved spatial-frequency discrimination and produced preliminary score-level motion/orientation signals. Those observations support continued acquisition; they do not yet establish a competent attentional observer or explain where failures arise.

Morgan, Albanna and Herman’s recurrent vision transformer provides a useful **functional comparison**, not an interchangeable mechanism. Their artificial-bias experiment forces spatial allocation toward a location and measures change reports and response timing; our KDA exposes signed associative matrices and emitted features, not the same allocation variable.[1] A local perturbation can test whether a representation causally influences behavior. It cannot, simply by being called “microstimulation,” establish correspondence to electrical current injection, a particular brain area, or the paper’s exact manipulation.

Main recommendation. First establish appropriate task competence and sham equivalence. Then perturb a calibrated local representation at matched sites and epochs, measuring psychometric changes, false reports, sensitivity and criterion. Treat enhancement, interference, disruption and null effects as distinct possible outcomes. Do not assume that increasing local activity should improve performance or reproduce the paper’s cue-versus-change timing effect.

Evidence boundary. This report describes the current joint lineage, not the historical orientation-only KDA trained with a different curriculum. Earlier successes are neither its initialization nor evidence that this learner has acquired the same mechanisms. No stimulation, inhibition, new calibration extraction or additional model evaluation was performed to write this document.

2. The implemented visual and memory hierarchy

The input is a float32 tensor $X \in [0, 1]^{B \times T \times 3 \times 100 \times 100}$. A batch contains one task and one condition, avoiding artificial sequence padding across different conditions. The model subtracts 0.5 and constructs a causal channel stack $[X_{t-2}, X_{t-1}, X_t]$. Unavailable history is zero in centered coordinates. This is explicit short-term sensory access, not a learned memory or an additional presented observation: two-frame tasks still contain exactly two observed frames. [C1, C3]

Spatial processing

Each encoder block is a stride-two convolution followed by eight-group GroupNorm and ReLU. The first kernel is 5×5 ; the remaining kernels are 3×3 , with padding preserving the specified downsampling convention. After each of the last three blocks, a learned 1×1 convolution projects current features to 32 channels for a distinct KDA module. Its 32-channel emission is concatenated with the block’s visual output.

Stage	Block input channels	Visual output	Output after memory concatenation
Conv 1	9	$32 \times 50 \times 50$	No KDA at this stage
Conv 2	32	$64 \times 25 \times 25$	$96 \times 25 \times 25$
Conv 3	96	$96 \times 13 \times 13$	$128 \times 13 \times 13$
Conv 4	128	$128 \times 7 \times 7$	$160 \times 7 \times 7$

The final concatenated map is flattened and passed through a learned linear projection and ReLU to 256 features. The configured optional feature normalization is the identity, not LayerNorm. A single-layer, 256-unit GRU receives the feature sequence, and the task’s linear head reads its final hidden state. No intermediate task decision is emitted by the deployed model. [C1]

What is stored, and where?

For each resolution s , spatial site p and associative head h , KDA stores

$$S_{t,p,h}^{(s)} \in \mathbb{R}^{8 \times 16}, \quad h \in \{1, 2\}.$$

The full state tensor is $B \times H_s \times W_s \times 2 \times 8 \times 16$. It is a learned key-to-value association at each feature location—not an eight-object store, sixteen named stimulus attributes or a biological synaptic count. Memory matrices and values can be signed. Each scale has its own trainable parameters and its own state. The two heads’ emissions form the 32-channel readout field. [C2]

Spatial coordinates are preserved in KDA, but a feature-map site is not an isolated image pixel or one experimental stimulus. Overlapping receptive fields, successive downsampling and convolutional mixing matter. GroupNorm also uses spatial statistics within channel groups, so the consequences of a localized feature intervention need not remain strictly local. Masks must be defined in scene coordinates, mapped to each scale, and verified against actual propagation.

The final GRU supplies another recurrent route. A cue could influence a later decision through local KDA, global GRU, their interaction or residual stacked input. The existence of spatial state does not prove that behavior uses it, and a global memory reset would confound these routes rather than isolate a local mechanism.

Verified size. There are **2,750,324 trainable parameters**: encoder blocks 256,992; projections 9,312; KDA modules 74,262; dense feature projection 2,007,296; GRU 394,752; task heads 7,710. KDA holds **215,808 state scalars per example** (160,000 / 43,264 / 12,544 across scales), or 863,232 fp32 bytes. Including the GRU gives 216,064 recurrent scalars, excluding the raw-frame stack, BPTT intermediates and optimizer storage. These are activation counts, not additional learned parameters. [R3]

3. KDA equations, retention and the readout bottleneck

At a fixed scale/site/head, suppress those indices and let $u_t \in \mathbb{R}^{32}$ denote projected current features. A learned 3×3 convolution produces packed query, key, value and gate vectors. Each head uses dimensions $8 + 8 + 16 + 8 + 1$. Queries and keys are L2-normalized with numerical epsilon 10^{-6} ; values are unconstrained. Elementwise sigmoid gives row-retention $\alpha_t \in (0, 1)^8$ and scalar write strength $\beta_t \in (0, 1)$. [C2]

The implemented update is

$$\begin{aligned}\bar{S}_t &= D_{\alpha_t} S_{t-1}, & e_t &= v_t - \bar{S}_t^\top k_t, \\ S_t &= \bar{S}_t + \beta_t k_t e_t^\top, & o_t &= S_t^\top q_t.\end{aligned}$$

The two 16-dimensional outputs are concatenated and transformed by a learned 1×1 convolution. That emitted field is joined to the current visual features before the next processing stage. The output projection has a bias; the recurrent matrix is not identical to the emitted feature field.

Interpretation and limited stability claim

This is a **prediction-error write**, not an unweighted running sum. The current key first retrieves a prediction; the update corrects its error toward the incoming value. Retention can forget old associations, and nonorthogonal keys can interfere. The same learned rule can implement useful persistence, selective replacement or an unhelpful recency bias depending on learned representations and gates.

Equivalently,

$$S_t = (I - \beta_t k_t k_t^\top) D_{\alpha_t} S_{t-1} + \beta_t k_t v_t^\top.$$

For a fixed gate/key sequence with $\|k_t\|_2 \leq 1$, the homogeneous state-transition norm is bounded by $\max_j \alpha_{t,j} \leq 1$. This follows from the eigenvalues of the rank-one correction and diagonal retention. It is a conditional algebraic bound, **not** proof of stable gradients or bounded activity in the entire learned recurrent network: values are unbounded, gates are input-dependent, upstream features depend on earlier memory emissions, and the global GRU forms an additional route. No forgetting time constant is inferred from this inequality.

Gate weights were initialized so retention starts at 0.9 and write strength at 0.5, initially independent of input. They remain trainable; those initialization values are not the trained retention law. Numerical checks reject nonfinite loss, gradients or parameters. This run does not clip gradients. [C2, C4]

Decision route and temporal interpretation

All spatial emissions must pass through a dense 256-dimensional projection and the shared GRU before reaching a task head. The projection preserves the possibility of location-specific weights through flattening, but creates a substantial compression relative to the full spatial state. A failed decision could therefore reflect poor encoding, weak retention, interference, inadequate comparison, or ineffective use of information that remains stored. Scores alone do not identify which explanation applies.

The GRU and KDA states reset at each trial; only trained parameters persist between trials. During a nominal blank, the frame stack can still contain a sample for the first two updates. A retention intervention must distinguish those residual sensory-access frames from genuinely stimulus-free processing. The ring task's final stack contains both sample frames: success there alone would not establish blank-delay memory. Frame counts are not biological milliseconds without a renderer-defined clock.

4. Task semantics and optimization exposure

The suite has thirteen task heads and 35 primary conditions. Task identity selects the head externally; the network is not learning which task the experimenter requested. Only images enter the sensory/recurrent path. Cue positions, renderer phases, labels, actual angles and image identities remain analysis metadata. [C3]

Family	Required report	Primary cells
Motion direction	Cardinal direction between two dot frames	1
Signed orientation	Sign of two-frame axial-angle change	1
Contrast	Which frame has greater contrast?	1
Spatial frequency	Which frame has higher frequency?	1
Chromatic increment	Which frame contains the defined colour increment?	1
Contour grouping	Which frame contains the aligned contour?	1
Natural spectral detail	Which image has greater high-frequency detail?	1
Ring orientation	Sign of change at ring-cued Gabor	1
Cued orientation	Target change matches spatial +/- instruction	4
Cued motion duration	Most frequent direction at the cued patch	4
Krauzlis change	Target change, versus foil change or catch	3
Spatial binding	Queried location participated in an orientation swap	4
Scene recognition	Probe is an exact member of the study set	12

The seven sensory tasks have two presented frames. Ring orientation has cue, two sample frames and probe, with no inserted blanks. Cued orientation, duration and binding use inserted delays $D \in \{0, 4, 12, 24\}$. Binding uses a retrospective query, not an encoding precue. Krauzlis conditions $B \in \{12, 20, 28\}$ vary baseline motion transitions, not memory delays. Recognition combines study lengths $N \in \{0, 4, 12, 24\}$ with three, four or five repetitions of the same probe. Repeated frames are not independent episodes. [C3]

Why these distinctions matter for causal comparison

The signed cued-orientation task is not an unrestricted “any change” detector. An opposite-sign target rotation is a negative report, even though a physical change occurred. Binding negatives still contain a foil swap. Krauzlis foil changes are deliberately negative. By contrast, a real uncued change is a valid detection event in the reference paper’s any-change task.[1] An intervention that increases reporting of uncued changes can therefore improve one task while violating another; calling both outcomes an attentional benefit would be misleading.

Training contract

All components train jointly from a fresh initial model, using mean cross-entropy and Adam at 10^{-4} with one parameter group, zero weight decay, no clipping, fp32 and full-sequence BPTT. Effective batch size is 32 via microbatches of four. A shuffled cycle contains one update per task; condition queues cycle within each task. A task head receives gradients only on that task’s updates, while shared processing receives all task losses. [C4]

The completed run used 715 updates, 22,880 episodes and 55 task-specific updates per head. The current eight-hour continuation preserves its weights, Adam state, streams and RNG, targeting 2,145 additional updates: 2,860 cumulative updates and 91,520 episodes if completed. The old horizon is not a plateau criterion. Training allocation, stimuli and architecture did not change during this continuation; checkpoint-selection policy changed prospectively, as discussed on page 6.

5. Preliminary behavioral metrics

Evidence frozen at 06:48:15 UTC, 23 September 2026. The running model had reached step 2647 / 84,704 episodes; its latest completed validation was step 2314. The table contrasts completed step-715 tests with that validation—not paired draws or the unmeasured performance of step 2647. V3 final tests were pending at the cutoff. BA averages eligible conditions equally within each task. [R1–R3]

Task	Completed 715 test BA	Interim 2314 val. BA	Interim val. AUC
Motion direction	26.56%	31.25%	0.722
Signed orientation	56.25%	50.00%	0.749
Contrast	100.00%	100.00%	1.000
Spatial frequency	63.28%	90.63%	0.981
Chromatic increment	99.22%	100.00%	1.000
Contour grouping	52.34%	51.56%	0.587
Natural spectral detail	92.97%	96.88%	0.997
Ring-cued orientation	46.09%	43.75%	0.520
Spatially cued orientation	50.59%	50.00%	0.498
Cued motion duration	24.41%	24.61%	0.516
Krauzlis target/foil change	50.00%	50.00%	0.441
Spatial orientation binding	51.76%	47.66%	0.483
Scene recognition, nonempty	50.00%	53.99%	0.524

Chance BA is 25% for the two four-direction tasks and 50% for the binary tasks. AUC chance is 0.5, including the macro one-versus-rest multiclass AUC used here. Ordinary validation cells contain 64 episodes, Krauzlis cells 100; the completed final test used 128 and 200 respectively. Small deviations around chance are not evidence of reliable competence. Reused validation draws and one training seed do not provide a population-general confidence statement.

What was acquired, and what remains unresolved?

Contrast, chromatic increments and image spectral detail show strong discrimination. These are low-level visual distinctions, not semantic recognition or a demonstration of maintained object identity. Spatial frequency shows substantial additional acquisition in validation. Motion direction's ranking AUC is now above its chance level, but its categorical decisions remain weak. Signed orientation likewise has an encouraging ranking score without improved decision accuracy. A threshold/criterion or score-scale mismatch is one possible contributor, not a demonstrated readout rescue; no post-hoc threshold was fit on these data.

At validation 2314, signed orientation predicted positive on all 64 trials (32 per true class): its AUC of 0.749 is not 75% classification accuracy. Contour grouping, cued judgments, binding and nonempty recognition remain weak. This does not locate the failure in memory rather than encoding, cue use, integration or readout. [R3]

Degenerate responses must remain visible

Krauzlis remained all-positive at validation 2314: each baseline had 57/57 target hits, 29/29 foil false reports and 14/14 catch false positives. The completed step-715 test showed the same degeneracy with twice those denominators. Empty recognition sets had 100% specificity in both snapshots; this must not inflate a nonempty recognition-learning claim. [R1, R3]

Full per-cell records, confusion matrices, difficulty strata and denominators remain in the saved JSON artifacts. The table summarizes all task families without treating favorable sensory averages as evidence that spatial working memory has been acquired.

6. Learning curves, selection and inference limits

The relevant question is whether continued unchanged training produces acquisition, not whether a larger step number alone certifies convergence. The first run stopped at a fixed exposure. It did not establish an asymptote, and each head had received only 55 updates. Its completed test performance is evidence about that exposure, not the model class’s capacity. [R1]

Prospective selection versus retrospective storytelling

The first selection rule maximized the worst chance-normalized task BA, using equal-task mean AUC only as a tie-breaker. It selected step 117 rather than 715: a noisy weak-task floor outweighed considerable sensory improvement. The official historical winner remains 117. Terminal 715 was retained as the latest training state and was explicitly authorized as the continuation parent; it was not relabeled as the old validation-selected model.

The new run prospectively selects by equal-task mean validation AUC, then mean chance-normalized BA, with earlier ties. Every condition and task remains visible. This acquisition-oriented criterion still does not guarantee balanced competence: strong tasks can raise an average while a difficult family remains at chance. Report the complete vector of task scores alongside the criterion, and do not choose a stimulation checkpoint because it gives the desired intervention effect. [C4]

Curve interpretation

Mean validation AUC was 0.630 at the step-715 baseline, 0.679 at step 1781 and 0.694 at step 2314. The sensory group’s mean AUC at 2314 was 0.862, whereas the five spatial/sequence tasks averaged 0.493. Thus progress is real but concentrated. A task’s AUC can improve before its decision accuracy, as in the orientation and cardinal-motion snapshots. Conversely, a constant action can yield an apparently stable BA while providing no useful event selectivity. [R2]

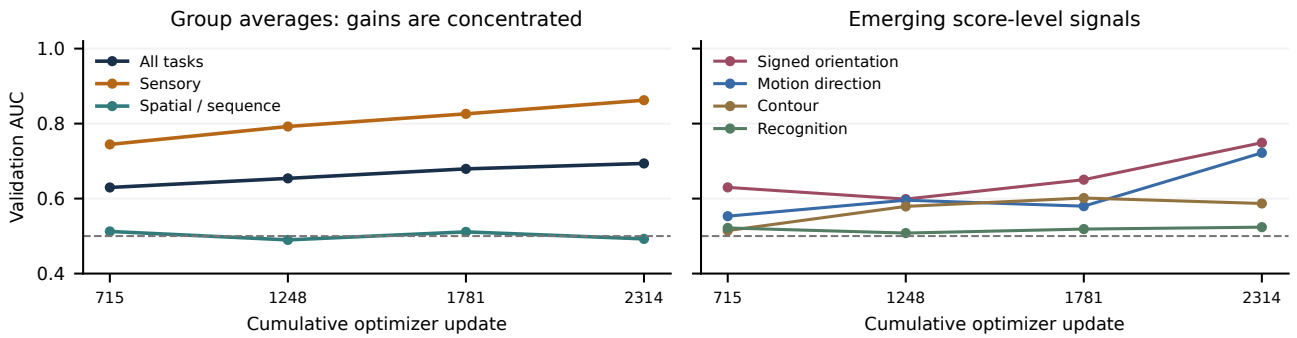


Figure 2: Saved validation snapshots only; dashed line is chance AUC. Group means are equal-task averages. No confidence intervals or newly generated model responses are implied.

What these measurements do not establish

No causal memory localization. We have not shown that the spatial matrices, rather than the GRU or frame stack, mediate successful decisions in this joint learner. Architecture determines available routes, not their behavioral use.

No comparable reaction time. Decisions are read at a prescribed final frame. Wall-clock inference time, GRU confidence, an arbitrary threshold crossing or an intermediate diagnostic head is not the paper’s learned wait/declare-change policy. An action-timing extension would be a separate model/task experiment.

No independent-source confidence interval. New episode seeds do not create new source photograph identities, and evaluation at trained centers is not novel-location generalization. Per-task curves should establish continued acquisition or a plateau before motivating architecture changes. At weak baseline performance, interventions chiefly test information use or vulnerability—not established attentional benefits.

7. What Morgan, Albanna and Herman actually manipulated

The pinned reference is *A recurrent vision transformer shows signatures of primate visual attention*, arXiv:2502.10955v1 (2025). Its recurrent ViT combines visual patches and activated memory to allocate attention, updates patch-based LSTM memory, and uses an actor–critic agent trained through reward. The task includes four Gabor locations, probabilistic cue validity and a wait/declare-change action policy. These are material differences from our end-of-trial supervised classifiers.[1]

Figure 5 is a computational allocation intervention

Section 4.3 and Figure 5 describe artificially increasing bias toward a spatial location, S1 or S4. The caption states that $\alpha_i(t_{\text{change}}) = 1$ means the transmission $Z(t_{\text{change}})$ is completely biased toward the corresponding $\xi_i(t_{\text{change}})$. It reports response-rate and reaction-time curves against orientation change, with 500 trials per plotted data point.[1] This is the paper’s computational analogue of causal manipulation in attention-related primate circuitry. It is not a literal model of electrode geometry, current amplitude or neural recruitment.

The reference attention is source-normalized per query ($\sum_j \alpha_{ij} = 1$; §6.5, Eqs. 7/14). Forcing one source to one would zero the others **if that normalization is preserved**, but Figure 5 does not specify the row/head overwrite or renormalization algorithm. The direct sensory residual remains (Eqs. 8/15); this is not erasure of all nonselected input.[1] Its spatial α is not our **per-key-row retention** α .

Read the legends by location, not colour. Blue means $\alpha_1 = 1$ in A/B/D/E but $\alpha_4 = 1$ in C/F. Bias toward the changing location helps; bias toward the cued location does not universally help. In C/F, the cue is at S1 and the change at S4: forcing S1 impairs reporting, while forcing S4 improves it.[1]

Section 4.3 reports weaker cue-time effects and context-dependent sensitivity/criterion effects.[1] These are textual claims here: the pinned PDF refers to supplemental panels X/16 that it does not contain. Exact clamp duration is also incompletely specified. We do not reconstruct missing numerical results or infer that a different recurrent architecture must share the timing effect.

Comparability map

Dimension	Reference recurrent ViT	Current joint KDA learner
Manipulated substrate	Spatially biased transmission	Local state, emission or feature drive
Memory organization	Patch-based LSTM with memory-guided attention	Multi-resolution local associative fields plus global GRU
Task report	Detect any real orientation change	Task-specific signs, target events, swaps, membership
Decision timing	Learned wait/declare-change policy	Fixed final-frame classification
Cue manipulation	Explicit validity levels	Task-defining cues; no matched validity series
Training	Sparse reward / actor–critic	Joint supervised cross-entropy
Tested causal result	Artificial bias, behavioral timing and detection	Not yet measured on this joint learner

The paper also reports that supervised variants could perform the task without reproducing the same cueing effect.[1] Consequently, even eventual high accuracy on our suite would not by itself predict primate-like attention dynamics. Functional comparability requires matched behavioral questions and controls, not merely recurrent state and a spatial mask.

8. A concrete computational microstimulation protocol

Proposed only. Freeze a preselected trained checkpoint and all model weights. Replay identical existing-task episodes under sham and intervention, preserving input rasters, labels, trial order and RNG. An analysis wrapper may use actual renderer phase indices to apply a pulse; those indices must not enter the deployed model or become a learned gate. First verify the unmodified wrapper against the ordinary forward path within a stated numerical tolerance.

Primary intervention: perturb the local emitted representation

At one scale, after KDA’s output projection and before concatenation, apply

$$\tilde{O}_t(p) = O_t(p) + \lambda g(t) m_s(p) \sigma_s d_s.$$

Start at the 13×13 scale. Here $m_s(p)$ is a scene-centred Gaussian mask with saved radius and unit peak; $g(t)$ specifies onset/duration; d_s is a unit-norm 32-channel direction. Set $\sigma_s^2 = \mathbb{E}_{\text{cal}} \|O - \mu_{\text{cal}}\|_2^2$, estimated at the declared task/epoch on independent calibration trials. Save achieved perturbation norms; equal peak doses do not imply equal total dose across mask sizes. Dose λ is dimensionless, not microamperes.

Calibrate a task-relevant direction independently of final evaluation: for example, a signed feature contrast from matched calibration trials, or a prespecified projection of a fitted calibration decoder. Validate the direction on a separate split. Include equal-norm random or orthogonal patterns and both signs where meaningful. Adding a positive constant to every signed channel is not an identified physiological excitation. Initial exploratory doses such as $0, \pm 0.25, \pm 0.5, \pm 1$ are **proposed relative scales**, not known safe or effective operating points.

This intervention changes what downstream processing receives at that timestep. It does not directly overwrite the same module’s state. It can nevertheless change higher-scale memory and the GRU, creating lasting behavioral consequences. Persistence after an emission pulse must not be misreported as proof that the directly stimulated local matrix was modified.

Secondary intervention: persistently perturb the association

To distinguish stored-content effects, perturb the updated state before readout:

$$\tilde{S}_t(p, h) = S_t(p, h) + \lambda g(t) m_s(p) \sigma_{S,s,h} a_h b_h^\top,$$

with calibrated unit vectors $a_h \in \mathbb{R}^8$, $b_h \in \mathbb{R}^{16}$. Specify one or both heads. The perturbed matrix must produce the current readout **and be carried forward**. Before output projection, its immediate effect is $\Delta o = \lambda g m \sigma_S (a_h^\top q_t) b_h$: even a large state pulse can be invisible to an orthogonal query. A readout-only copy instead tests transient access. This perturbs associative content, not isolated attention allocation or identified biological synapses.

Controls and closer—but nonidentical—allocation analogues

A local emission gain, $\tilde{O} = (1 + \lambda g m) O$, changes existing influence without adding a new pattern. A norm-matched suppression toward a stated reference tests a complementary direction. Neither creates the reference ViT’s competitive spatial allocation: increasing one site does not automatically remove other sites’ contributions. A renormalized routing operator could impose that competition, but it would be a new diagnostic computation, not an existing KDA control variable or an exact replication. Do not silently substitute it for a state or emission intervention.

Log substrate, scale, head, site mask, epoch, dose, achieved feature/state change and downstream propagation. Start with one declared scale and substrate rather than a broad search for an attractive result; confirmatory doses/sites must be fixed before their final evaluation.

9. Experimental design and behavioral readouts

A meaningful microstimulation study compares **where**, **when** and **what** was perturbed while holding physical evidence fixed. Use paired episodes, not unrelated stimulus draws, for sham-versus-pulse contrasts. Save per-trial responses and the actual spatial arrays; region averages cannot reconstruct per-timestep maps. No extraction or intervention run is authorized by this document.

Site × epoch × dose design

Compare the cued/queried site, a matched uncued site and an off-target control. For retrospective binding, the queried location is not known from an encoding cue; distinguish that analysis label from information available to the model. Match masks by scene-space geometry and disclose unequal feature coverage across resolutions. Spatially localized injection can have distributed consequences through convolutions and GroupNorm.

Separate cue presentation, sample encoding, early nominal blanks, genuinely stimulus-free retention and probe/change/report. The three-frame stack carries samples into the first two blanks, so use actual input-stack contents to verify blank eligibility. Cue pulses may be duplicated through the raw input stack if applied to images, whereas a one-step internal emission pulse is a different intervention. Declare the timing convention explicitly. Use trial-relative discrete frames unless a renderer supplies physical timing; do not equate all task frames with the Krauzlis 100 Hz clock.

Psychometrics before aggregate accuracy

For orientation, plot positive-report probability against **signed target change**, conditioning on foil changes, cue meaning, stimulated site and dose. For the ring task, the label is rotation sign; for signed cued orientation, it is agreement with the instructed sign, with unchanged negatives. A positive response at zero change has different semantics in those tasks. Do not mechanically fit the reference paper’s change-detection psychometric function to a different report rule.

For Krauzlis, plot target hit rate, foil-event positive rate and catch positive rate separately. For binding, a foil-only exchange is the negative condition, not a no-change image. Recognition requires nonempty membership sensitivity plus separate empty-set specificity. Motion-duration effects should be stratified by count-winner margin and recent motion history where the native metadata permits, rather than assuming a change-detection interpretation.

Where binary signal/noise classes are meaningful, estimate

$$d' = \Phi^{-1}(H) - \Phi^{-1}(F), \quad c = -\frac{1}{2}[\Phi^{-1}(H) + \Phi^{-1}(F)].$$

Define H and F for each task; analyze Krauzlis catch and foil comparisons separately rather than silently pooling them. Correct rates of zero or one with a declared finite-sample rule, for example $(k + 0.5)/(n + 1)$. For four-choice tasks use the confusion matrix and class-specific effects, not an unexplained binary d' . Criterion shifts alone can change response rates without enhancing discrimination.

Uncertainty, specificity and falsification

Estimate within-episode sham/pulse differences with paired resampling; keep multiple doses, delays or repetitions of the same base episode grouped. Photograph-based uncertainty should account for reused source identities. Independent trained seeds address a different uncertainty than more trials from one model. Fix primary comparisons in advance and label secondary site/epoch searches exploratory.

A selective site-by-epoch interaction, a graded dose relation and task-appropriate psychometric shifts would support a functional effect. Uniform degradation at all sites, positive-report increases with equal false-alarm increases, or effects duplicated by random patterns favor nonspecific disruption or response bias. A null effect can reflect ineffective stimulation, saturation, redundant pathways or poor baseline competence; verify the achieved manipulation before concluding that a region is unnecessary.

10. Expected outcomes, comparability and next decisions

Are the experiments comparable? At the level of a causal question—does increasing the influence of a spatial representation change perception in a site- and time-specific manner?—yes. At the level of manipulated variable, task, action policy and biological implementation, not directly. A rigorous study should report which levels were matched rather than collapse them into a yes/no equivalence claim.

Do we expect the same results? Not as a default. The reference’s change-time effect and weak cue-time effect motivate directional hypotheses.[1] Our cue encoding, temporal stacking, local associative state and supervised decision rule could produce strong encoding effects, persistent storage distortions, immediate report bias, compensatory processing or no measurable benefit. The present weak spatial competence makes a clean attentional-enhancement prediction especially premature.

Observation in a future study	Supported interpretation	Not established
Stimulated-site sensitivity rises with stable false alarms	Selective improvement under that intervention	Same circuit as FEF/SC or identical ViT mechanism
Hits and false alarms rise together	Response-bias component	Improved sensory information
Local state pulse persists across true blanks	Perturbed recurrent route affects later behavior	All memory resides at that site
Emission pulse changes behavior but state pulse does not	Substrate-dependent causal access at tested doses	Stored information is absent
Cue pulse differs from probe pulse	Temporal specificity	Replication unless task/timing are matched
All interventions cause broad degradation	Possible nonspecific disruption	Attention-specific causal mechanism

Decision sequence. Finish the authorized training and inspect per-task acquisition. Select the checkpoint without stimulation-test outcomes. Establish the relevant baseline psychometrics, separate calibration/selection/test episodes, verify sham equivalence, and only then run one bounded intervention family. If the existing task cannot support a valid-neutral comparison or genuine response timing, label those outcomes unavailable; implementing the reference task or a timed action policy would require a separately authorized experiment. Do not change teaching or architecture merely to obtain the anticipated effect.

Biological anchor. Cavanaugh, Alvarez and Wurtz stimulated superior colliculus around a possible dot-motion change: 600 ms beginning 150 ms before the first motion epoch ended (Fig. 2A). Hits generally increased while false positives changed little, motivating explicit false-report controls rather than hits alone.[2] That electrical intervention recruits biological circuitry; neither a ViT allocation clamp nor a KDA feature pulse has a validated conversion to its current, tissue extent or temporal scale.

Reproducibility and source map

[C1] WorkingMemory/PlainBaseline/accum.py:40–75: convolutional hierarchy, stack, projection, GRU and heads. [C2] PreAttentiveVision/TemporalIntegration/accumulators.py:19–66: exact KDA update and gates. [C3] SecondPass/TaskSuite/{README.md,catalog.json,suite.py}: task laws, streams and reporting. [C4] SecondPass/JointTraining/{worker.py,core.py,continuation_v3.py,AMENDMENT_V3.md}: optimizer, selection and continuation contract.

[R1] SecondPass/JointTraining/runs/fresh_kda_joint_01_continuation_v2/: report.json, test_terminal.json, test_selected.json. Terminal checkpoint 715 was used for the completed-test column, not historical winner 117. [R2] ../fresh_kda_joint_01_continuation_v3_8h/: scheduled validation JSONs and progress.jsonl. Snapshot files and figure-building scripts are retained beside this report. Historical orientation-lineage interventions are not evidence for this joint model.

[R3] Beside this report: metrics_snapshot.json, architecture_metrics_notes.md, paper_comparison_notes.md and paper_evidence/verified_quotes.json. The audit verifies six complete 13-task/35-cell evaluations, parameter counts, checkpoint shapes and source digests without model inference.

Sources

[1] <https://arxiv.org/html/2502.10955v1> — Morgan, Albanna and Herman (2025), A recurrent vision transformer shows signatures of primate visual attention, pinned v1

[2] <https://www.jneurosci.org/content/26/44/11347> — Cavanaugh, Alvarez and Wurtz (2006), Enhanced Performance with Brain Stimulation: Attentional Shift or Visual Cue?, Journal of Neuroscience 26:11347–11358